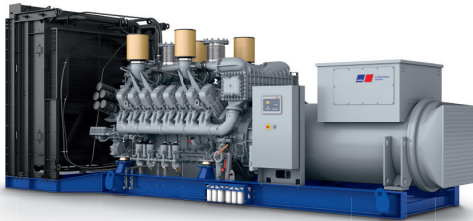




Diesel Generator Set

mtu 16V4000 DS2250

380V – 11 kV/50 Hz/standby power/fuel consumption optimized
16V4000G74F/water charge air cooling



Optional equipment and finishing shown. Standard may vary.

Product highlights

Benefits

- Low fuel consumption
- Optimized system integration ability
- High reliability
- High availability of power
- Long maintenance intervals

Support

- Global product support offered

Standards

- Engine-generator set is designed and manufactured in facilities certified to standards ISO 2008:9001 and ISO 2004:14001
- Generator set complies to ISO 8528
- Generator meets NEMA MG1, BS 5000, ISO, DIN EN and IEC standards
- NFPA 110

Power rating

- System ratings: 2260 kVA - 2370 kVA
- Accepts rated load in one step per NFPA 110*
- Generator set complies to G3 according to ISO 8528-5
- Generator set exceeds load steps according to ISO 8528-5*

Performance assurance certification (PAC)

- Engine-generator set tested to ISO 8528-5 for transient response
- 85% load factor
- Verified product design, quality and performance integrity
- All engine systems are prototype and factory tested

Complete range of accessories available

- Control panel
- Power panel
- Circuit breaker/power distribution
- Fuel system
- Fuel connections with shut-off valve mounted to base frame
- Starting/charging system
- Exhaust system
- Mechanical and electrical driven radiators
- Medium and oversized voltage alternators

Emissions

- Fuel consumption optimized

Certifications

- CE certification option
- Unit certificate acc. to VDE-AR-N 4110

* Changes to the standard parameter sets (alternator-regulator and genset-controller) are necessary



A Rolls-Royce
solution

Application data ¹⁾

Engine			Liquid capacity (lubrication)	
Manufacturer	mtu		Total oil system capacity: l	300
Model	16V4000G74F		Engine jacket water capacity: l	175
Type	4-cycle		Intercooler coolant capacity: l	50
Arrangement	16V		Combustion air requirements	
Displacement: l	76.3			
Bore: mm	170			
Stroke: mm	210		Combustion air volume: m³/s	2.3
Compression ratio	16.4		Max. air intake restriction: mbar	50
Rated speed: rpm	1500		Cooling/radiator system	
Engine governor	ECU 9			
Max power: kWm	1965			
Air cleaner	dry		Coolant flow rate (HT circuit): m3/hr	68.5
Fuel system			Coolant flow rate (LT circuit): m3/hr	30
			Heat rejection to coolant: kW	730
			Heat radiated to charge air cooling: kW	320
			Heat radiated to ambient: kW	90
			Fan power for electr. radiator (40°C): kW	38
Maximum fuel lift: m	5		Exhaust system	
Total fuel flow: l/min	20			
Fuel consumption ²⁾				
	l/hr	g/kwh	Exhaust gas temp. (after turbocharger): °C	485
At 100% of power rating:	452.23	191	Exhaust gas volume: m³/s	5.8
At 75% of power rating:	342.7	193	Maximum allowable back pressure: mbar	85
At 50% of power rating:	240.3	203	Minimum allowable back pressure: mbar	30

Standard and optional features

System ratings (kW/kVA)

Generator model	Voltage	fuel consumption optimized					
		without radiator			with mechanical radiator		
		kWel	kVA*	AMPS	kWel	kVA*	AMPS
Leroy Somer LSA52.3 S7 (Low voltage Leroy Somer standard)	380 V	1864	2330	3540	1840	2300	3494
	400 V	1864	2330	3363	1840	2300	3320
	415 V	1864	2330	3242	1840	2300	3200
Leroy Somer LSA52.3 L12 (Low voltage Leroy Somer oversized)	380 V	1864	2330	3540	1840	2300	3494
	400 V	1864	2330	3363	1840	2300	3320
	415 V	1864	2330	3242	1840	2300	3200
Marathon 744RSL7092 (Low voltage Marathon)	380 V	1848	2310	3510	1808	2260	3434
	400 V	1848	2310	3334	1808	2260	3262
	415 V	1808	2260	3144	1808	2260	3144
Marathon 1020FDL7093 (Low voltage Marathon oversized)	380 V	1848	2310	3510	1808	2260	3434
	400 V	1848	2310	3334	1808	2260	3262
	415 V	1808	2260	3144	1808	2260	3144
Marathon 1020FDL7093 (Low voltage Marathon engine output optimized)	380 V	1896	2370	3601	1856	2320	3525
	400 V	1864	2330	3363	1824	2280	3291
	415 V	1864	2330	3242	1824	2280	3172
Marathon 1020FDH7097 (Medium volt. marathon)	11kV	1864	2330	122	1808	2260	119
Leroy Somer LSA53.2 XL9 (Medium volt. Leroy Somer)	11 kV	1880	2350	123	1840	2300	121

* cos phi = 0.8

1 All data refers only to the engine and is based on ISO standard conditions (25°C and 100m above sea level).
2 Values referenced are in accordance with ISO 3046-1. Conversion calculated with fuel density of 0.83 g/ml. All fuel consumption values refer to rated engine power.